THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** re-examine assumptions, models or measurements	mismatch drives further exploration
2	**(a)** spheres joined by bonds	a model for structure
3	**(c)** pounds and kilograms	unit mix-up in density
4	**(c)** solution	uniform composition
5	**(d)** milk diluted with water	colloids scatter light
6	**(b)** solid carbon dioxide that sublimes	used for storing ice cream
7	**(c)** dinitrogen tetroxide	prefixes show the number of atoms; second element ends in -ide
8	**(a)** 44 g	mass is conserved
9	**(c)** 58.5 u	23 + 35.5
10	**(c)** 3	electrons lost, gained or shared to complete the octet
11	**(b)** protons and neutrons	nucleons
12	**(c)** 2, 8, 8, 1	fill shells in order with maximum 2, 8, 8, 2

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	The choice is made on purpose to keep the problem simple enough while still answering the question (1); extra details can be added later for accuracy (1).	1+1
14	$\frac{5}{100} \times 100 = 5\%$ m/v.	1+1
15	By distillation (1); acetone boils at about 56 °C, water at 100 °C, so acetone vaporises first and is condensed (1).	1+1
16	Reactants $= 6.92$ g; products $= 1.76 + 0.72 + 4.44 = 6.92$ g; equal, hence verified (1+1).	1+1
17	Cathode rays were the same irrespective of the cathode material or the gas used, so electrons are a basic component of all atoms (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Ask what happens during an eclipse (the Moon blocks sunlight) (1); does temperature or any chemical/biological change in food occur? measure it (1); no mechanism or evidence is found, so the claim is unsupported (1).	1+1+1
19	Filtering removes insoluble impurities (1.5); the acid prevents unwanted reactions and gives pure crystals (1.5).	1½+1½
20	Weigh both solutions together before mixing (1); on mixing a white precipitate of barium sulfate forms with sodium chloride (1); the total mass remains unchanged — the law is verified (1).	1+1+1
21	It predicted that α-particles would pass through or be deflected slightly (1); observed: some deflected sharply, a few rebounded (1); positive charge is concentrated, not spread (1).	1+1+1
22	Positive charge and most mass concentrated in a tiny nucleus; electrons revolve around it like planets; most of the atom is empty (1+1+1).	1+1+1
	OR: $0.1\ \text{mm} = 10^{-4}$ m (1); atoms $= 10^{-4}/10^{-10}$ (1) $= 10^{6}$ (about a million) (1).	1+1+1
23	Theories are tested against new evidence (1); when predictions fail, scientists change assumptions or models (1); this openness to correction lets science improve (1).	1+1+1
	OR: Ground crew used density in pounds per litre instead of kilograms per litre (1); fuel was about 15 000 litres short (1); using SI units everywhere avoids conversion errors (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

24	(i) Solution: salt in water (clear, no settling); colloid: milk, fog (scatter light, stable); suspension: muddy water (settles) (3). (ii) Scattering of light by colloidal particles (1). (iii) Pass a torch beam through a dilute milk sample; the beam path is visible (1).	3+1+1
	OR: (i) Separation of mixture components based on different speeds on a paper (1). (ii) 3 cm strip, pencil line 2 cm from the bottom, ink spot at centre, dip in water with the spot above the water level (2). (iii) Water rises carrying the dyes; different colours separate at different heights as they interact differently with paper and solvent (1). (iv) Otherwise the spot would dissolve into the solvent instead of travelling (1).	1+2+1+1
25	(i) Mass of helium is twice what protons alone give (1). (ii) James Chadwick in 1932 discovered the neutral neutron (1). (iii) Protons and neutrons (1). (iv) Neutrons reduce repulsion between protons and strengthen the nuclear force (2).	1+1+1+2
	OR: (i) 17 p, 18 n, 17 e (1). (ii) 2, 8, 7 (1). (iii) Valency 1; chlorine (1). (iv) It gains one electron to form Cl^- (anion) (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** how molecules are formed	an explanation based on evidence
2	**(c)** check whether an answer is reasonable	estimation builds intuition and detects errors
3	**(a)** is backed by careful testing and evidence	evidence-based
4	**(a)** paper chromatography	different components move at different speeds
5	**(b)** a saturated solution	definition
6	**(c)** naphthalene	solid → vapour directly
7	**(a)** covalent bond	shared electron pair
8	**(c)** a double bond	two shared pairs
9	**(b)** $CaCl_2$	Ca^{2+} and 2 Cl^-
10	**(d)** 10^5 (one lakh)	10^{-10} m versus 10^{-15} m
11	**(c)** 18	neutrons = 35 − 17
12	**(a)** James Chadwick	1932

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	The fuel was miscalculated; the aircraft was about 15 000 litres short and had to glide to an emergency landing (2).	2
14	A solution that cannot dissolve any more solute at that temperature (2).	2
15	Carbon 6 g needs 16 g oxygen, so $6 + 16 = 22$ g of CO_2 (1+1).	1+1
16	In a chemical reaction matter can neither be created nor destroyed; total mass of reactants equals total mass of products (2).	2
17	Most passed straight through (1); most of the atom is empty space (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Keep: distance, average speed, stops at crossings (1+1). Ignore: colour of cycle, school name, tyre brand (1)—they do not change the time.	2+1
19	Mass $= \frac{4}{100} \times 300 = 12$ g (3).	1+1+1
20	(i) Atoms are neither created nor destroyed, only rearranged, so mass is conserved (1.5). (ii) Atoms combine in fixed simple ratios, giving fixed mass ratios (1.5).	1½+1½
21	$10^{-10}/10^{-15} = 10^5$ times (1); if the atom were a 100 m cricket ground, the nucleus would be a black pepper grain (1) at the centre (1).	1+1+1
22	Questions on humidity (above 80% last time?), wind speed and direction, temperature trend (1 each) — they use data and past patterns, not just ‘dark clouds’.	1+1+1
	OR: Real systems are too complex to study in full detail (1). A model keeps only the key factors, e.g. falling object: ignore air resistance to study gravity (1). Assumptions simplify; they are added back for accuracy later (1).	1+1+1
23	Distillation: acetone vaporises first and is condensed (2). Evaporation would let both vaporise and lose the liquid (1).	2+1
	OR: $\frac{5}{100} \times 200 = 10$ mL of glacial acetic acid (2), made up to 200 mL with water (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Pressure raises the boiling point of water (1). (ii) Cooking changes food chemically; nutrients/microbes are affected (1). (iii) Relationships between pressure, temperature and time can be expressed and tested using equations (2). (iv) Real problems cross boundaries made by humans to organise knowledge (1).	1+1+2+1

	OR: (i) A unit conversion error (1). (ii) It had about 15 000 litres less fuel than required (1). (iii) Standard units everywhere avoid conversions and mistakes (2). (iv) Mass and volume (or density and fuel quantity) (1).	1+1+2+1
25	(i) 71 g (2). (ii) 117 g (1). (iii) 46 g (1). (iv) Law of Constant Proportions (1).	2+1+1+1
	OR: (i) Atoms; indivisible; identical for an element; different for different elements; simple whole-number ratios; constant relative number (3). (ii) Atoms are conserved ⇒ mass conserved; simple ratios ⇒ fixed ratio by mass (1). (iii) Atoms are divisible into subatomic particles (electrons, protons, neutrons) (1).	3+1+1

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** a simplified way of looking at a real system that keeps only what matters for the question	models ignore details on purpose
2	**(c)** careful observation	as explained in the chapter
3	**(c)** Predictions that fail lead to better ideas	self-correcting nature
4	**(b)** Tyndall effect	light path becomes visible
5	**(b)** 20 %	$\frac{20}{100} \times 100 = 20$
6	**(c)** crystallization	differences in solubility with temperature
7	**(c)** Cl^- ion	chlorine gains one electron
8	**(a)** Na_2SO_4	2 Na^+ and SO_4^{2-}
9	**(d)** 2 g and 16 g	1 : 8 by mass
10	**(a)** streams of negatively charged particles	electrons
11	**(b)** 6	neutrons = 12 − 6
12	**(a)** Leucippus and Democritus	ancient Greece

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	In science a theory is an explanation based on careful testing and critical examination (1). It may change with new evidence but it is not an untested guess (1).	1+1
14	Solubility of solids generally increases with temperature and that of gases decreases, so the value depends on temperature (2).	2
15	In the open set-up the CO_2 produced escaped; in the closed set-up it stayed inside, so the total mass was unchanged (2).	2
16	A helium atom has 2 protons but mass about 4 times that of hydrogen, so uncharged particles must be present (1); Chadwick discovered the neutron (1).	1+1
17	The nucleus is small, dense and positively charged containing most of the mass; electrons revolve around it; most of the atom is empty (1+1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Breaths per minute ≈ 14 (1); breaths per day ≈ 20 000 (1); volume per breath ≈ 0.5 litre ⇒ about 10 000 litres per day (1).	1+1+1
19	A method for separating components whose boiling points differ by less than about 25 °C using a fractionating column (1.5); used for crude petroleum into petroleum gas, petrol, kerosene, diesel (1.5).	1½+1½
20	$8 : 2 = 4 : 1$, the same ratio (2). So yes — the compound has a fixed ratio irrespective of how it is prepared (1).	2+1
21	Berzelius proposed using Latin names for international use (1); Fe (ferrum), Na (natrium), Hg (hydrargyrum) (1); first letter capital, second small (1).	1+1+1
22	Mixture heated in a flask; lower-boiling liquid vaporises (1); vapour passes through a condenser and condenses (1); distillate collected separately; used to recover solvent/extract fragrance (1).	1+1+1
	OR: Solubility (g per 100 g water) against temperature (1). Inferences: B dissolves much more in hot water than cold (1); cooling a hot saturated solution of B deposits a lot of crystals (1).	1+1+1
23	S : O = 40 : 60 = 2 : 3 (1); oxygen $= \frac{3}{2} \times 20 = 30$ g (2).	1+2
	OR: $\frac{4}{3} \times 9 = 12$ g (3).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

24	(i) B (the curve is higher) (1). (ii) B, because its solubility drops sharply on cooling (2). (iii) B, because large differences in solubility with temperature give a better yield (2).	1+2+2
	OR: (i) Too little does not protect the crop and too much can harm it (1). (ii) $\frac{1}{100} \times 100 = 1\%$ v/v (1). (iii) 20 L = 20000 mL; 1% = 200 mL (2). (iv) A homogeneous mixture (solution) (1).	1+1+2+1
25	(i) Maximum $2n^2$ per shell; outermost shell ≤ 8; filled stepwise K, L, M, N (2). (ii) 2,8,2; 2,8,7; 2,8,6; 2,8,8,1; 2,8,8,2 (2.5). (iii) 2, 1, 2, 1, 2 (½).	2+2½+½
	OR: (i) Gas at low pressure in a glass tube with two electrodes and high voltage; rays from cathode deflected in electric and magnetic fields (2). (ii) Cathode rays are streams of negatively charged electrons, independent of the cathode material and the gas, so electrons are present in all atoms (1). (iii) A positive sphere with embedded electrons; atoms are neutral (2).	2+1+2

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